

ISA CERTIFIED PERSONAL TRAINER COURSE

WEEK 9 – NUTRITION + PROGRAM
DESIGN 6 (SPECIAL POPULATIONS)

- CHAPTER 6
NUTRITION FOR HEALTH AND
FITNESS
- CHAPTER 12
CONSIDERATIONS FOR CLIENTS
WITH OBESITY



CONTENT COVERED

Part I: Nutrition for Health and Fitness

- ☐ Nutrition for Health and Fitness: Introduction
- ☐ Nutrition and Scope of Practice
- ☐ Dietary Reference Intakes (DRI)
- ☐ Acceptable Macronutrient Distribution Ranges (AMDR)
- ☐ Fueling before, during, and after exercise
- ☐ Fluid and Hydration before, during, and after exercise
- ☐ The Dietary Guidelines
- ☐ Eating Disorders
- ☐ Exercise vs. Nutrition

Part II: Considerations for Clients with Obesity

- ☐ Considerations for Clients with Obesity: Introduction
- ☐ Definition and Causes of Obesity
- ☐ Exercise and Weight Management
- ☐ Nutritional Guidelines for Obesity
- ☐ Exercise Guidelines for Obesity – Cardiorespiratory Training
- ☐ Exercise Guidelines for Obesity – Muscular Training



CHAPTER 6

NUTRITION FOR HEALTH AND FITNESS

NUTRITION FOR HEALTH AND FITNESS: INTRODUCTION

Lesson Objectives



Explain what is within a personal trainer's scope of practice as it relates to nutrition

Discuss the current Dietary Guidelines and other evidence-based eating patterns



Explain proper fueling and hydration principles for before, during, and after exercise



NUTRITION AND SCOPE OF PRACTICE



SCOPE OF PRACTICE

- Share general non-medical nutrition information
- Guidelines endorsed by the U.S. Federal Government, Dietary Guidelines for Americans (www.health.gov/dietaryguidelines) and the MyPlate recommendations at (www.ChooseMyPlate.gov)
- Complete continuing education to develop nutrition competency
- Develop relationships with registered dietitians (RDs)



COLLABORATING WITH REGISTERED DIETICIANS

- Refer clients them to an R.D. if needed
- Eating disorders, client requests for meal planning, a need for detailed nutrition information, or supplement recommendation etc.



NUTRITIONAL SUPPLEMENTS

- Recommending supplements is not allowed
- Discuss safety and efficacy of supplements.
- Provide clients with evidence-based resources
- Take a conservative approach when discussing supplements

DIETARY REFERENCE INTAKES (DRI)

Dietary Reference Intakes (DRI)

EAR	Estimated Average Requirement: Adequate intake in 50% of an age-gender specific group
RDA	Recommended Dietary Allowance: Level of intake of a nutrient that is adequate to meet the known needs of all healthy persons
AI	Adequate Intake: Recommended nutrient intake level that appears to be sufficient for good health
UL	Tolerable Upper Intake Level: Maximal intake that is unlikely to pose a risk to adverse health effects to almost all individuals
AMDR	Acceptable Macronutrient Distribution Ranges: A range of intake specified as a percentage of total energy intake. Used for sources of energy, such as fats and carbohydrates.

ACCEPTABLE MACRONUTRIENT DISTRIBUTION RANGES (AMDR)

	Source of Goal ¹	Female 14-18	Male 14-18	Female 19-30	Male 19-30	Female 31-50	Male 31-50	Female 51+	Male 51+
Calorie level(s) assessed		1,800	2,200, 2,800, 3,200	2,000	2,400, 2,600, 3,000	1,800	2,200	1,600	2,000
Macronutrients									
Protein, g	RDA	46	52	46	56	46	56	46	56
Protein, % kcal	AMDR	10-30	10-30	10-35	10-35	10-35	10-35	10-35	10-35
Carbohydrate, g	RDA	130	130	130	130	130	130	130	130
Carbohydrate, % kcal	AMDR	45-65	45-65	45-65	45-65	45-65	45-65	45-65	45-65
Dietary fiber, g	14g/1,000 kcal	25.2	30.8	28.0	33.6	25.2	30.8	22.4	28.0
Added sugars, % kcal	DGA	<10%	<10%	<10%	<10%	<10%	<10%	<10%	<10%
Total fat, % kcal	AMDR	25-35	25-35	20-35	20-35	20-35	20-35	20-35	20-35
Saturated fat, % kcal	DGA	<10%	<10%	<10%	<10%	<10%	<10%	<10%	<10%
Linoleic acid, g	AI	11	16	12	17	12	17	11	14
Linolenic acid, g	AI	1.1	1.6	1.1	1.6	1.1	1.6	1.1	1.6

* Refer to <http://health.gov/dietaryguidelines/2015/guidelines/appendix-7/> for information about the needs for children and adolescents.

¹ RDA = Recommended Dietary Allowance; AMDR = Acceptable Macronutrient Distribution Range; DGA = 2015-2020 Dietary Guidelines recommended limit; 14 g fiber per 1,000 kcal = basis for AI for fiber; AI = Adequate Intake

Reprinted from United States Department of Agriculture (2015). 2015-2020 Dietary Guidelines for Americans (8th ed.). www.health.gov/dietaryguidelines

Table 6-1: Daily Nutritional Goals for Age-Gender Groups based on Dietary Reference Intakes and Dietary Recommendations

FUELING BEFORE, DURING, AND AFTER EXERCISE

Pre-Exercise Fueling

- Carb loading – eat a meal 4-6 hours prior to the workout
- Eating a small carb-and-protein snack 30-60mins before exercise increases glucose availability
- Pre-exercise snack should be high in carbs; low in fat and fiber, moderate in protein; approximately 400-800 calories

Fueling during Exercise

- Athletes should consume glucose-containing snacks and consume 30-60g of carbs/hour of training
- Consume in small amounts in 15-20min intervals
- Adding protein to carbs during exercise helps improve performance

Post-Exercise Replenishment

- Post-workout meals include mostly carbs with some protein
- Carb intake of 1.5g/kg of body weight in the first 30mins after exercise, then every 2 hours for 4-6 hours
- Refueling should begin within 30mins after exercise and followed by high-carb meal within 2 hours

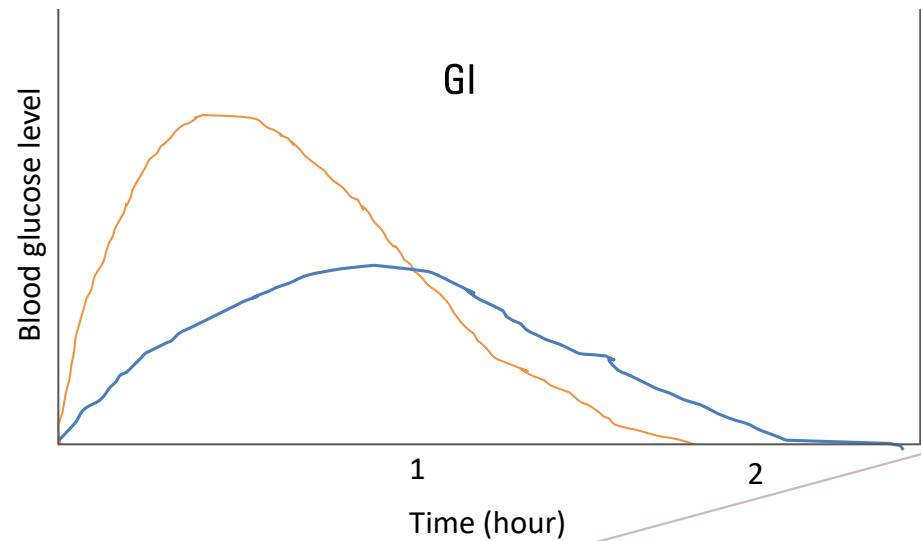
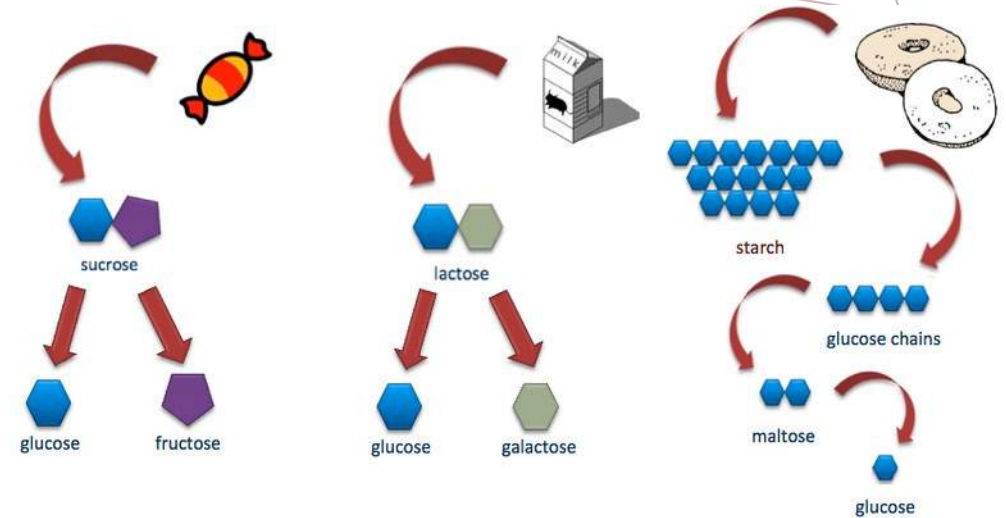


CARBOHYDRATES

- Functions:

- 1) Provide energy: 4 kcal/g
- 2) Protein sparing
- 3) Oxidation of fat
- 4) Stored energy

- Recommendations for Intake
- Glycemic Index (GI) & Glycemic Load (GL)



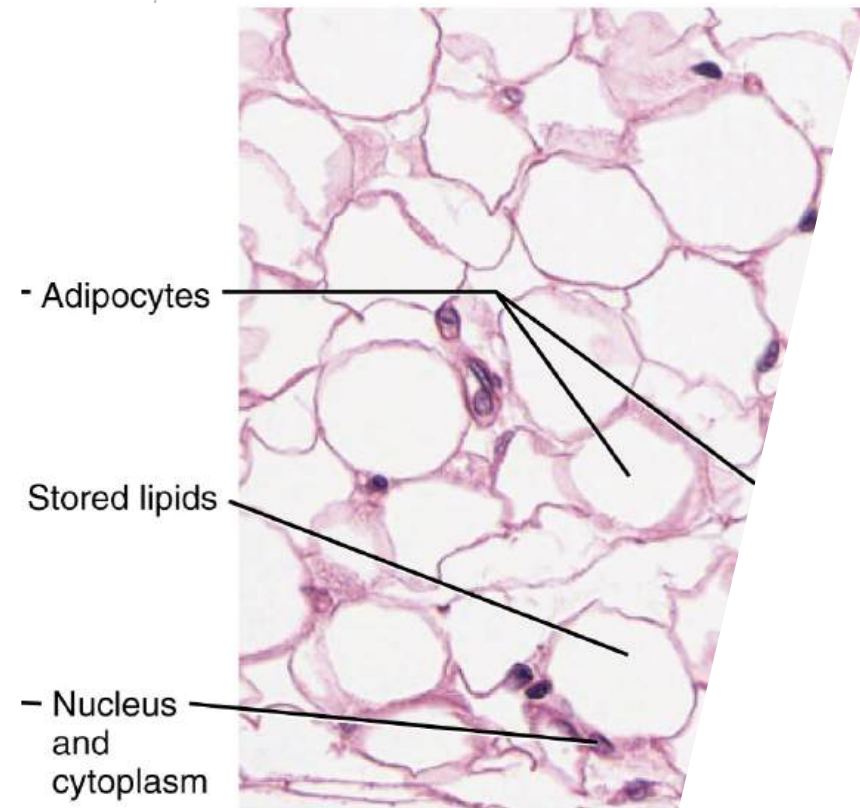


PROTEIN

- Amino acids chained through peptide bonds
- Forms brain, nervous tissue, blood, muscles, skin, teeth, hair, muscles
- Transport mechanism for iron, vitamins, minerals, fats, and O₂
- Acid–base fluid balance
- Recommended intake
- Protein quality
 - Evaluated by the protein digestibility corrected amino acid score (PDCAAS)
- 4 kcal/g

FAT

- Protection, insulation, cell structure, nerve transmission, vitamin absorption, and hormone production
- Hormones
- Carries fat soluble vitamins
- Omega-3 and omega-6 are (polyunsaturated) fatty acids
- 9 kcal/g



FUELING BEFORE, DURING, AND AFTER EXERCISE

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FLUID AND HYDRATION BEFORE, DURING, AND AFTER EXERCISE

	Fluid	Comments
Before Exercise	▶ Drink 5–7 mL/kg (0.08–0.11 oz/lb) at least 4 hours before exercise (12–17 ounces for a 154-lb individual)	▶ If urine is not produced or is very dark, drink another 3–5 mL/kg (0.05–0.08 oz/lb) two hours before exercise. ▶ Sodium-containing beverages or salted snacks will help retain fluid.
During Exercise	▶ Monitor individual body-weight changes during exercise to estimate sweat loss. ▶ Composition of fluid should include 20–30 mEq/L of sodium, 2–5 mEq/L of potassium, and 5–10% of carbohydrate.	▶ Prevent a >2% loss in body weight. ▶ Amount and rate of fluid replacement depends on individual sweating rate, environment, and exercise duration.
After Exercise	▶ Consumption of normal meals and beverages will restore euhydration. ▶ If rapid recovery is needed, drink 1.5 L/kg (23 oz/lb) of body weight lost.	▶ Goal is to fully replace fluid and electrolyte deficits. ▶ Consuming sodium will help recovery by stimulating thirst and fluid retention.

Reprinted with permission from American College of Sports Medicine (2018). ACSM's Guidelines for Exercise Testing and Prescription (10th ed.). Philadelphia: Wolters Kluwer.

Table 6-4: Fluid Replacement Recommendations Before, During and After Exercise



DIETARY GUIDELINES

The 2015-2020 Dietary Guidelines for Americans offers 5 key recommendations for good nutrition:

1. Follow a healthy eating pattern across the lifespan
2. Focus on variety, nutrient density, and amount
3. Limit calories added from sugars and saturated fat, and to reduce sodium intake
4. Shift to healthier food and beverage choice
5. Support healthy eating patterns for all



DIETARY GUIDELINES

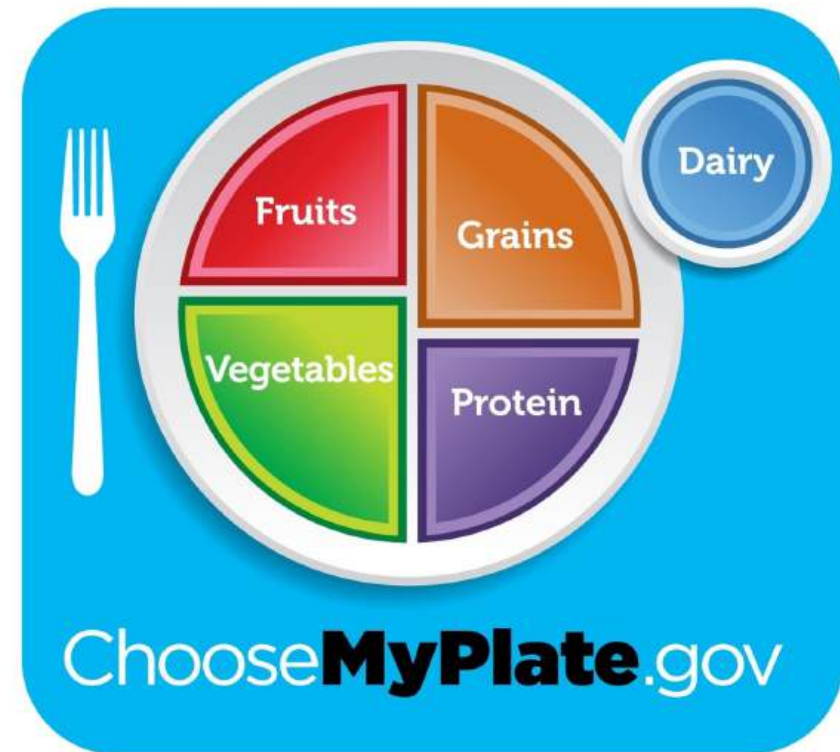
Key Guideline 1: *Follow a healthy eating pattern across the lifespan*

- ❖ Choose a healthy eating pattern to maintain a healthy body weight and support nutrient adequacy
- ❖ Main components of a healthy eating pattern include:
 - A variety of vegetables and fruits
 - Grains
 - Fat-free or low-fat dairy
 - A variety of foods rich in protein
 - Limiting the amount of saturated fat and trans fat
 - Limiting sodium to <2300mg/day
 - Limiting alcohol to 1 drink/day for women and 2 drinks/day for men



DIETARY GUIDELINES - MYPLATE

- ❑ Implementation of the Dietary Guidelines through MyPlate
- ❑ A dinner plate divided into 4 sections – fruits, vegetables, protein, and grains, accompanied by a glass of non-fat milk
- ❑ Can visit www.ChooseMyPlate.gov to identify daily caloric and nutrient needs



DIETARY GUIDELINES

Key Guideline 2: *Focus on variety, nutrient density, and amount*

- ❖ Choose an appropriate amount of food from each of the 6 groups of nutrient-dense foods:
 - Vegetables
 - Fruits
 - Grains
 - Dairy
 - Protein foods
 - Oil

Choose a variety of nutrient-dense foods from each food group in recommended amounts.

Example Meal:



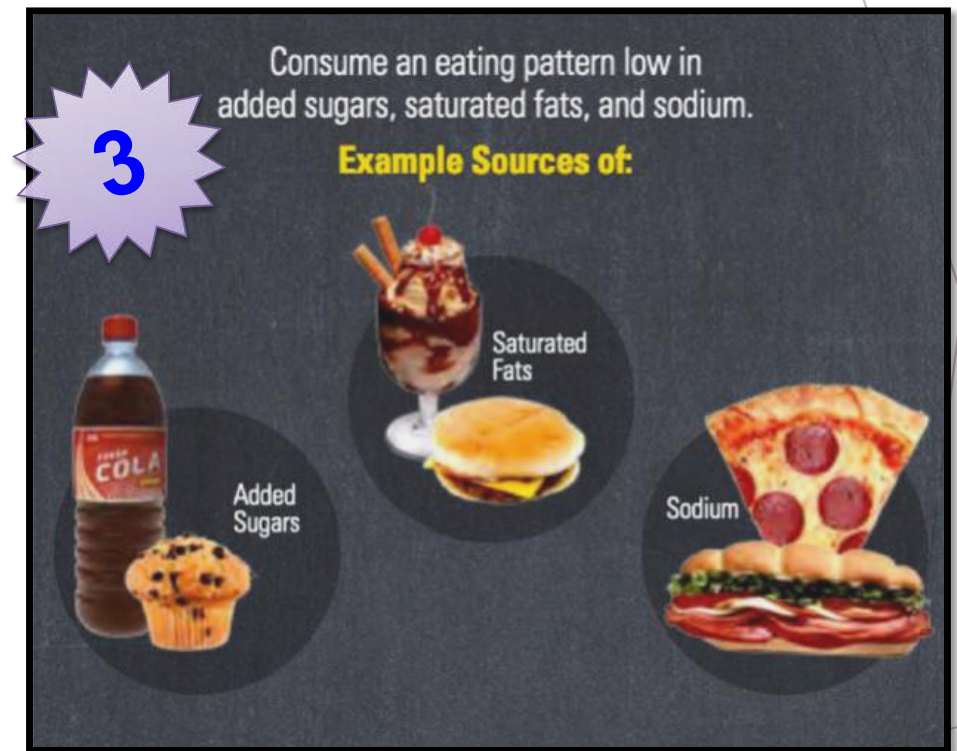
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Lettuce & Celery	Vegetables	Grains	Whole-Grain Bread
Apples & Grapes	Fruits	Dairy	Fat-Free Milk
Chicken Breast & Unsalted Walnuts	Protein	Oils	Mayonnaise

DIETARY GUIDELINES

Key Guideline 3: *Limit calories added from sugars and saturated fat, and to reduce sodium intake*

- ❖ Adopt an eating pattern that is low in added sugars, saturated fat, and sodium
- ❖ Consume <10% of total calories from added sugars
- ❖ Consume a diet containing <10% of total calories from saturated fat
- ❖ Replace with foods with polyunsaturated and monounsaturated fat
- ❖ Consume as little trans fat as possible
- ❖ Reduce sodium to <2300mg/day
- ❖ Individuals with hypertension are advised to follow the DASH (Dietary Approaches to Stop Hypertension) eating plan



DIETARY GUIDELINES

Key Guideline 4: *Shift to healthier food and beverage choices*

- ❖ Consume nutrient-dense foods and beverages across all food groups
- ❖ Consider cultural and personal preferences
- ❖ Some of these recommendations:
 - Shift from high calorie snacks to nutrient-dense snacks
 - Shift from snacks high in sodium to unsalted snacks
 - Shift from solid fats to oils
 - Shift from beverages with added sugar to no-sugar-added beverages

4

Replace typical food and beverage choices with more nutrient-dense options. Be sure to consider personal preferences to maintain shifts over time.

Example:



Meal A

➔

Shift

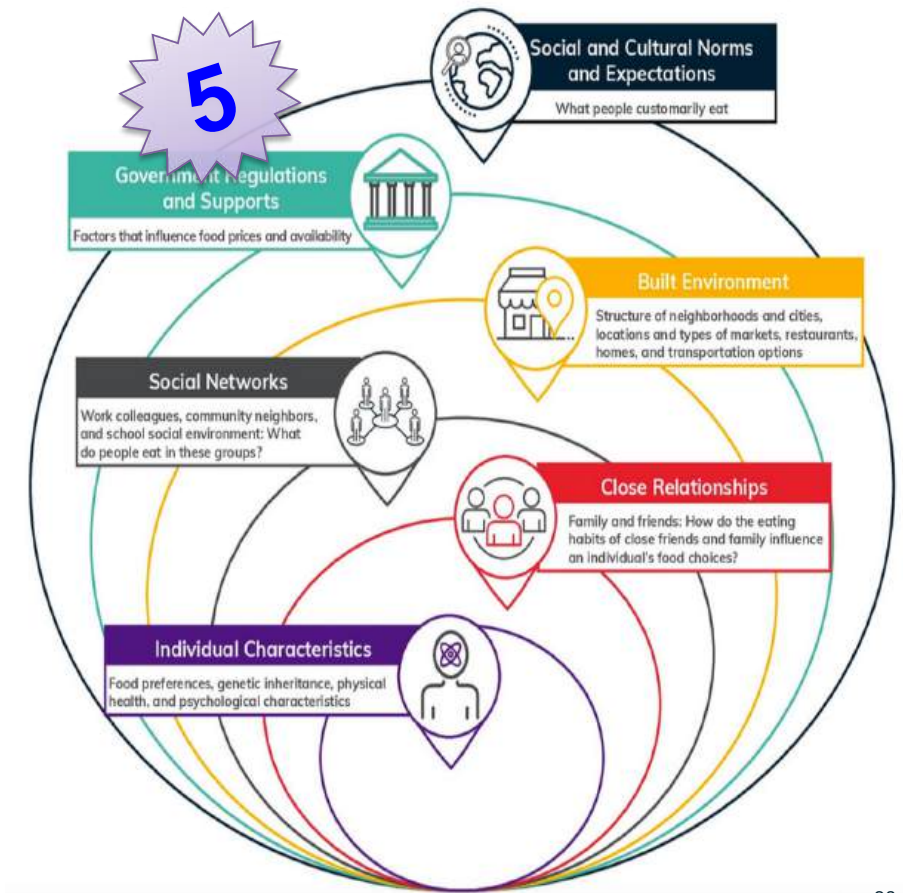


Meal B

DIETARY GUIDELINES

Key Guideline 5: *Support healthy eating patterns for all*

- ❖ A person's diet is influenced by external factors. Understood via the socio-ecological model - society has a role to play in supporting healthy choices for individuals
- ❖ PTs can contextualize individual choices and motivators by expanding out through the different socio-ecological levels
- ❖ Intervene using a combination of strategies to impact behavioral change
- ❖ Interventions should extend beyond traditional education on nutrition - should help build skills, reshape the environment, and reestablish social norms to facilitate healthy choices



Original Label

Nutrition Facts	
Serving Size 2/3 cup (55g)	
Servings Per Container 8	
Amount Per Serving	
Calories 230	Calories from Fat 72
% Daily Value*	
Total Fat 8g	12%
Saturated Fat 1g	5%
Trans Fat 0g	
Cholesterol 0mg	0%
Sodium 160mg	7%
Total Carbohydrate 37g	12%
Dietary Fiber 4g	16%
Sugars 12g	
Protein 3g	
Vitamin A	10%
Vitamin C	8%
Calcium	20%
Iron	45%
* Percent Daily Values are based on a 2,000 calorie diet. Your daily value may be higher or lower depending on your calorie needs.	
	Calories: 2,000 2,500
Total Fat	Less than 65g 80g
Sat Fat	Less than 20g 25g
Cholesterol	Less than 300mg 300mg
Sodium	Less than 2,400mg 2,400mg
Total Carbohydrate	300g 375g
Dietary Fiber	25g 30g

New Label

Nutrition Facts	
8 servings per container	
Serving size	2/3 cup (55g)
Amount per serving	
Calories	230
% Daily Value*	
Total Fat 8g	10%
Saturated Fat 1g	5%
Trans Fat 0g	
Cholesterol 0mg	0%
Sodium 160mg	7%
Total Carbohydrate 37g	13%
Dietary Fiber 4g	14%
Total Sugars 12g	
Includes 10g Added Sugars	20%
Protein 3g	
Vitamin D 2mcg	10%
Calcium 260mg	20%
Iron 8mg	45%
Potassium 235mg	6%
* The % Daily Value (DV) tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.	

EATING DISORDERS

Anorexia Nervosa

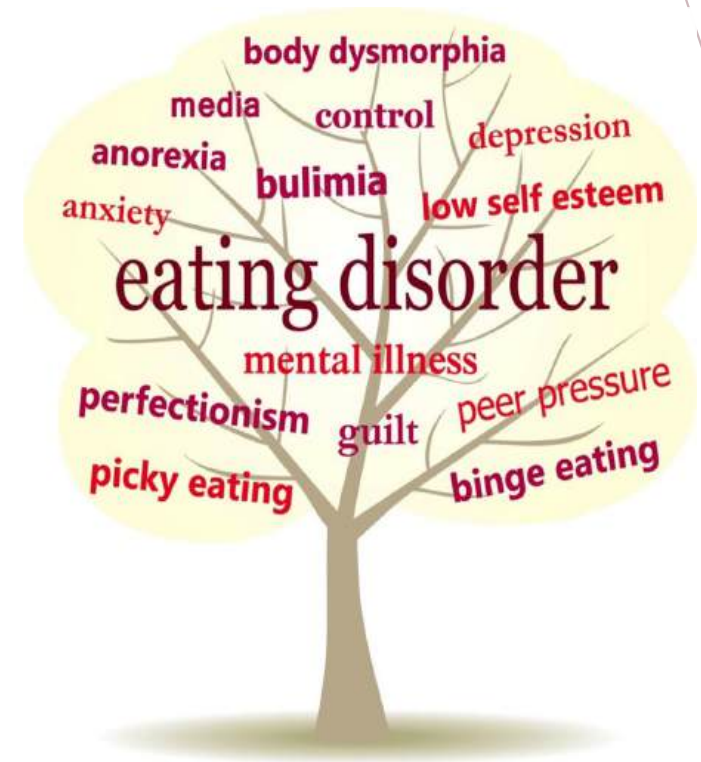
- An eating disorder characterized by refusal to maintain body weight
- Intense fear of gaining weight or becoming fat
- May practice excessive exercise, use laxatives and vomit to reduce weight (but to a lesser extent than Bulimia)
- Body image disturbances

Bulimia Nervosa

- An eating disorder characterized by recurrent episodes of uncontrolled binge eating and behavior such as self-induced vomiting
- May be repeated several times a week or even several times a day

Binge Eating Disorder

- An eating disorder characterized by frequent binge eating (without purging)
- Feelings of being out of control when eating



EXERCISE VS. NUTRITION



NUTRITION FOR HEALTH AND FITNESS: CONCLUSION

Nutritional Balance allows for

- ❑ Optimal functioning
- ❑ Prevention of chronic disease

Dietary Guidelines, AMDRs and
Recommendations for Exercise
Fuelling and Hydration serve as
basic nutritional foundation

Baseline for working with special populations clientele

CHAPTER 12
CONSIDERATIONS FOR
CLIENTS WITH OBESITY

CONSIDERATIONS FOR CLIENTS WITH OBESITY: INTRODUCTION

- | | |
|---|---|
| ❑ | Describe major factors that contribute to obesity |
| ❑ | Explain the major components of behavioral weight-loss plan |
| ❑ | Outline key dietary strategies for weight loss and weight maintenance |
| ❑ | Discuss the role of physical activity and exercise in prevention and treatment of obesity |

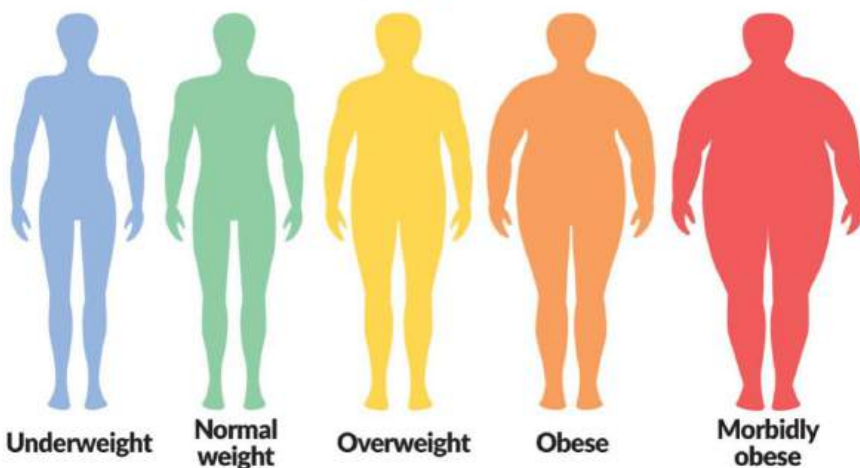


DEFINITION AND CAUSES OF OBESITY

When am I obese?

The easiest way to determine whether a person is overweight or obese is through measuring their body mass index (BMI).

$$\text{BMI} = \frac{\text{Weight in kg}}{(\text{Height in m})^2}$$



<18.5	18.5-24.9	25-29.9	30-39.9	40 and above
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Possible Causes of Obesity

[PG 557-561]



Environmental Factors



Stress



Genetics and Hormones



Sleep



Energy Balance

HORMONES PRODUCED BY FAT TISSUE

- Leptin
- Adiponectin
 - Facilitate cell uptake of blood glucose, increase insulin sensitivity, stimulate fat breakdown.
- Immune hormones
 - Fat tissue produces immune hormones such as tumor necrosis factor-alpha, interleukin-6, plasminogen activator inhibitor 1, angiotensin II, and other cytokines.
 - Cytokines
- Ghrelin
- Peptide YY

EXERCISE AND WEIGHT MANAGEMENT

A client-centered approach to working with clients with obesity

- ❖ Lifestyle interventions and increased physical activity should be the first options
- ❖ Utilize 3 components of behavioral weight loss program:
 1. Caloric reduction to produce a deficit of 500-1000kcal daily
 2. Increased physical activity
 3. Behavioral-change strategies
- ❖ Refer obese clients to RDs for a more personalized dietary plan
- ❖ Combination of these factors allows for better weight-management success



EXERCISE AND WEIGHT MANAGEMENT

Setting Body Weight Goals

- ❖ PTs should establish realistic and attainable weight loss goals.
- ❖ Maintaining current body weight can also be seen as a sign of progress.
- ❖ Pharmacological interventions and/or bariatric surgery should be a last resort and only for those with:
 - BMI 40 and above, or
 - BMI 35 and above with one obesity-related health issue.



NUTRITIONAL GUIDELINES FOR OBESITY



The ADA (American Dietetic Association, 2009) Weight Loss Guidelines

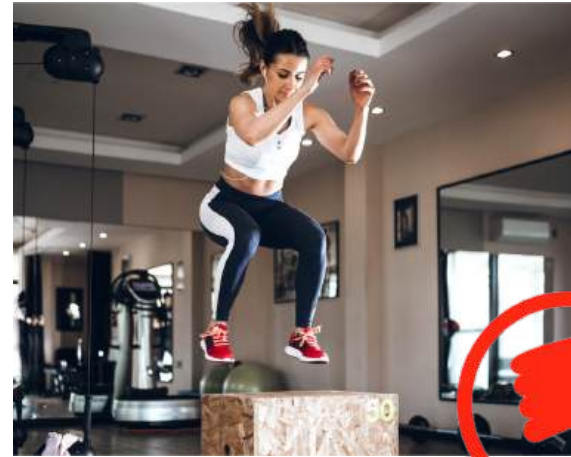
- ❖ Achieve a caloric deficit of 500-1000kcal/day, which produces a weight loss of 1-2lbs/week
 - Reduce EI by 250kcal and increase EE by 250kcal = 500kcal
- ❖ Practice portion control and reduce consumption of caloric-dense foods that are high in fat and sugar content
- ❖ Increase consumption of fruits and vegetables.
- ❖ Consume enough protein to preserve lean tissue

EXERCISE GUIDELINES FOR OBESITY

– CARDIORESPIRATORY TRAINING

2018 ACSM FIIT Guidelines (Frequency, Intensity, Time, Type)

- ❖ F: At least 5 days/week
- ❖ I: Moderate intensity. Best to exercise below VT_1 or at RPE 3-4
- ❖ T: 30-60min/day to achieve volume of 150-250min/week
- ❖ T: Low impact rhythmic exercises for large muscle groups (e.g. brisk walking or cycling)



EXERCISE GUIDELINES FOR OBESITY

– MUSCULAR TRAINING

2018 ACSM FIIT Guidelines (Frequency, Intensity, Time, Type)

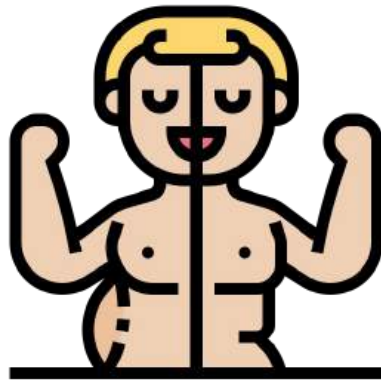
- ❖ F: 2-3 days/week
- ❖ I: Moderate-intensity for 8-12 reps for 2-4 sets for exercise
- ❖ T: Full body muscular training program takes 20-30mins to complete
- ❖ T: Work all major muscle groups using free weights and machines



CONSIDERATIONS FOR CLIENTS WITH OBESITY: CONCLUSION



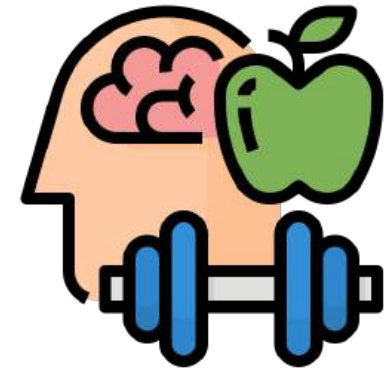
Realistically manage
client's expectations



Create exercise programs
that cater to physiological
demands of obesity



Boost their confidence and
enjoyment of physical activity



Focus on sustainable eating
changes, regular activity, social
support and self-compassion

CONCLUSION

Nutrition

Knowledge in Nutrition can be an effective add-on for developing an all-rounded program for obese clients, while also grasping the specifics of program design and modification for such clients.

